

Tanish Singh Rajpal

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Quantum information algorithms | quantum error correction | quantum chemistry | learning-based methods

EDUCATION

Carnegie Mellon University

M.S. in Information Technology - Information Security
Information Networking Institute

Pittsburgh, PA
Expected May 2027

NMIMS University

B.Tech. in Computer Engineering
GPA: 3.83/4.00 | Valedictorian, graduating class

Mumbai, India
May 2025

RESEARCH EXPERIENCE

Learning-based routing in heterogeneous quantum networks

May 2026 - present

Independent Study, CMU Information Networking Institute | Advisor: Prof. Michael Mattarock

- Built a decentralized PPO router in NetSquid and an end-to-end simulator that applies link and memory noise, entanglement swapping, and delayed classical corrections before measuring endpoint fidelity.
- Trained five policies on 20 x 20 grids and evaluated them without retraining on unseen 20 x 20 and 30 x 30 topologies. In a local, idealized 30 x 30 simulation, the learned policy achieved 61.2% request acceptance versus 64.3% for a negative-log-fidelity routing baseline.
- Manuscript and reproducibility checks in progress; current evaluation does not model network contention or throughput.

Differentiable quantum-chemical models (DFTBML3)

2026 - present

Independent Study, CMU Chemistry | Advisor: Prof. David Yaron

- Working on differentiable density-functional tight-binding models for chemical Hamiltonians, preserving physical structure while testing trainable components against established implementations. Results in progress.

A hybrid architecture for post-quantum EAP-TLS

Aug 2026 - present

Master's Thesis, CMU | Advisors: Prof. Pedro Bustamante and Prof. Michael Mattarock; Reader: Prof. Patrick Tague

- Investigating whether oversized ML-DSA certificates can cause Path MTU Discovery blackholes in WPA-Enterprise 802.1X authentication.
- Designing an ns-3 evaluation of an edge QUIC proxy and QKD-secured wired backbone, with hypotheses covering handshake latency, RADIUS compute, IPv4/IPv6 behavior, and key-delivery capacity. Evaluation in progress.

SELECTED QUANTUM PROJECTS

Phase Hunter - mapping quantum spin phases under noise

Sep 2026

Q-SITE Hacks 2026, Quantum Coalition Scientific Track runner-up | Team Popeyes

- With Paul Jeon and Sumedhaa Ruhil, built a playable phase-boundary search backed by reproducible PennyLane simulations. Compared clean, noisy, and error-mitigated phase maps under limited measurement budgets. [Code and notebook](#).

Quantum portfolio optimization using QAOA

Apr 2026

Team Lead, Yale YQuantum Hackathon | Third place

- Led the Markowitz-to-QUBO-to-Ising formulation and QAOA circuit design for an insurance portfolio optimization prototype. [Repository](#).

PEER-REVIEWED PUBLICATIONS

Journal article

Rajpal, T. S., and Naithani, A. (2026). "NeuroCrypt: A Neuro Symbolic AI Ecosystem for Advanced Cryptographic Data Security and Transmission." IEEE Transactions on Artificial Intelligence, 7(1), 512-521. [doi:10.1109/TAI.2025.3577605](https://doi.org/10.1109/TAI.2025.3577605)

Conference papers

Naithani, A., and Rajpal, T. S. (2025). "Blockchain-Based Digital Health Passport: A Decentralized Framework for Secure and Interoperable Health Data Management." IEEE ZINC 2025. [doi:10.1109/ZINC65316.2025.11100281](https://doi.org/10.1109/ZINC65316.2025.11100281)

Naithani, A., Rajpal, T. S., and Mistry, V. (2024). "TAV-Cart: Transaction Assistance and Validation Cart System for Effortless Shopping and Checkout." IEEE ICACCM 2024. [doi:10.1109/ICACCM61117.2024.11059075](https://doi.org/10.1109/ICACCM61117.2024.11059075)

Naithani, A., Jain, V., Rajpal, T. S., and Mistry, V. (2024). "AI Based Goalkeeper for Penalty Shot Prediction Using RNN and Pattern Matching Algorithm." IEEE ISDFS 2024. [doi:10.1109/ISDFS60797.2024.10527307](https://doi.org/10.1109/ISDFS60797.2024.10527307)

HONORS AND TRAINING

- Q-SITE Hacks 2026: runner-up, Quantum Coalition Open Challenge (Scientific Track); third place, IBM Quantum Coding Challenge; third place, Overall Coding Challenge; fourth place, IonQ Championship.
- IBM Qiskit Global Summer School 2026 - Quantum Excellence (Advanced). [Credential](#).
- Third place, Yale YQuantum Hackathon (2026); first place, Crypto Quest Cybersecurity Competition (2024).

RESEARCH METHODS AND TOOLS

Methods: reinforcement learning, PPO, quantum-network simulation, QAOA, QUBO/Ising formulations, experimental design.

Platforms: NetSquid, ns-3, PennyLane, Qiskit, Python, Git/GitHub.

Security and networking: EAP-TLS, ML-DSA, 802.1X, PMTUD, QKD, Burp Suite, Metasploit.

INDUSTRY EXPERIENCE

Vulnerability Assessment and Penetration Testing Intern

Oct 2023 - Jan 2024

Newton's Apple, Pune, India

- Identified and supported remediation of three critical vulnerabilities in a confidential Android application; performed static Java code analysis with the development team.

ACADEMIC SERVICE

Student Ambassador, CMU Information Networking Institute

Oct 2025 - present

Support prospective students and help plan community events.